

Curriculum Vitae
Frederick (Tzujie) Huang
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EDUCATION

B. S., <i>with Distinction</i>, in Biomedical Engineering	2025
<i>The University of North Carolina at Chapel Hill, Chapel Hill, NC</i>	
B. S., <i>magna cum laude</i>, in Biomedical Engineering	2025
<i>North Carolina State University, Raleigh, NC</i>	

PROFESSIONAL EXPERIENCE

Research Technician	2025-Present
<i>Department of Comprehensive Dentistry, School of Dentistry, UT Health San Antonio</i>	
Teaching Assistant	2024-2025
<i>Lampe Joint Department of Biomedical Engineering, UNC-Chapel Hill & NC State University</i>	
Project Intern	2024-2025
<i>Lampe Joint Department of Biomedical Engineering, UNC-Chapel Hill & NC State University</i>	
Clinical Engineering Intern	2024-2025
<i>Department of Physical Medicine and Rehabilitation, UNC School of Medicine</i>	
Research Assistant	2023-2024
<i>Eshelman School of Pharmacy, UNC-Chapel Hill</i>	
Research Assistant	2022-2023
<i>Department of Applied Physical Sciences, UNC-Chapel Hill</i>	

PROJECTS

Anki Remote Controller <i>PCB Design, DFM</i>	August 2025
- Built a 5 button wireless device for professional school students navigating Anki's review software. - Implemented and deployed 200+ lines of code for secure, rapid pairing with personal devices.	
Reaction Time Datalogger <i>C++, Serial Communications</i>	May 2025
Investigated reaction times (RTs) across audio, visual, and tactile stimuli, utilizing AVR family microcontrollers for signal acquisition and processing.	
Biospecimen QA/QC in Omic Processing <i>LC/MALDI-TOF, LipidSearch 5.0</i>	May 2025
- Studied biospecimen degradation during extraction processes, building SOPs for sample preparation. - Reduced lipid breakdown in biospecimen processing by up to 22.9% among 1740+ measured samples.	
Re-imaging Treatments for Pectus Carinatum <i>Medical Device Prototyping</i>	May 2025
- Developed a chest brace in collaboration with UNC Health PM&R for pectus carinatum patients to give providers real-time feedback following ISO 14971 and 13485 standards. - Altered patient Haller index by an average of 0.2, projecting a 30% decrease in treatment time.	
Non-Primary Weight Bearing Bone Development <i>HR-pQCT, Technical Writing</i>	April 2025
- Experimental proposal to map stress/strain distributions in the distal tibia and ulna for healthy, sedentary, pre-menopausal patients under non-cyclic loading using HR-pQCT based finite element modeling. - Projected real, cortical, and trabecular thickness of ulna to increase by 1.6, 37, and 4.7%.	
Scaffold Manufacturing for Valvular Cells <i>Technical Writing</i>	April 2025
Design proposal for an electrospun polycaprolactone scaffold manufactured for seeding with up to 3×10^9 valvular endothelial/interstitial cells.	
Sourcing Insulin Analogs from c. geographus <i>PyMOL, DeepMind, Chimera</i>	December 2024

- Researched venom properties of fish-hunting cone snails *in silico* to engineer rapid pharmacokinetic properties into a 57 amino acid long ultra stable single-chain insulin analog.
- Tested 15+ candidate molecules for suitable receptor interactions.

Developing Mouse Osteoblast Cell Lines | *Cell Culture and Passaging, BSL2* February 2024

- Cultured and passaged 7F2 mouse osteoblast cells lines for junior lab students.
- Sustained at least 92.6% cell viability across 8 batches containing at least 1×10^6 cells.

Finite Element Modeling in Biological Systems | *MATLAB* May 2024

- Modeled transient diffusion across drug-releasing patches and IL-6 releasing mammary duct platforms.
- Demonstrated as a proof-of-concept from Kutys-Polacheck et al. 2020.

Alcohol Oxidation through Light Mediated Reactions | *Organic Synthesis, HNMR* May 2023

- Synthesized low-cost and nontoxic pyrylium salt catalysts operating under mild conditions for drug discovery applications.
- Characterization with HNMR and cyclic voltammetry.

PRESENTATIONS

Poster (with S. Mathur, K.S. Paye, M.N. Cole, M.J. Gilbert, M.K. Hederick)

- "The Brace Check—Novel Data Collection for Monitoring Pectus Carinatum", Lampe Joint Department of Biomedical Engineering Senior Design Symposium, Raleigh, NC, April 2025.

Poster

- "Quality Management in Lipidomic Extractions to Develop a Lipidomic Integrity Number", UNC-Chapel Hill Undergraduate Research Symposium, Chapel Hill, NC, April 2025.

TEACHING EXPERIENCE

Course Name and Number	Enrollment	Type	Year
BMME 209 Materials Science of Biomaterials	37	Ugrad	2024-2025
Music Theory and Classical Performance	2	Tutor	2022-2024

SELECTED HONORS & AWARDS

IPEP Distinguished Scholar, <i>School of Medicine, University of North Carolina at Chapel Hill</i>	2025
Abrams Scholar, <i>College of Engineering, North Carolina State University</i>	2024
Dean's List, <i>University of North Carolina at Chapel Hill</i>	2021
AP Scholar with Distinction, <i>College Board</i>	2021

VOLUNTEER EXPERIENCE

Music Instructor	1/2022 - 1/2025
<i>Musical Empowerment @ UNC-Chapel Hill</i>	Chapel Hill, NC
• Led chapter efforts to teach one-on-one lessons for violin/viola to K-12 students from 250+ families in Chapel Hill-Carrboro School System.	
Executive Officer	5/2022 - 5/2025
<i>UNC Mens Ultimate Blue—Darkside</i>	Chapel Hill, NC
• Contracted sponsor agreements with local businesses and managed a \$60,000+ budget from 2023-2025. • Synthesized efforts with a local organization, Triangle Ultimate, and St. Augustine's University club sports to establish the US's first active HBCU ultimate frisbee team.	

TECHNICAL SKILLS

- Prototyping: Additive and Subtractive Manufacturing, CAD, Microcontrollers, PCB Design.
- Languages: MATLAB, Python, C++, L^AT_EX
- Software: PyMOL, ImageJ, Onshape, Fusion, AutoCAD, LipidSearch, BLAST, FASTA, AutoDock Vina, NI MAX, LabChart, Chimera, ClusPro, Office Suite
- Laboratory: Aseptic Technique, Cell Culture and Passaging, TEM, Sterilization